

GRONINGEN EELDE, Netherlands

WMO: 062800

Lat: 53.1236N

Lon: 6.5847E

Elev: 5

StdP: 101.26

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.6	(c) -5.6	(d) -10.7	(e) 1.5	(f) -6.5	(g) -8.2	(h) 1.9	(i) -4.8	(j) 14.4	(k) 8.5	(l) 13.1	(m) 8.8	(n) 3.4	(o) 80	(p) 0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.4	(c) 28.8	(d) 20.1	(e) 26.3	(f) 19.3	(g) 24.1	(h) 18.2	(i) 21.1	(j) 26.5	(k) 20.0	(l) 25.0	(m) 19.0	(n) 23.1	(o) 3.8	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.3	14.1	23.4	18.3	13.2	22.0	17.4	12.4	20.9	61.1	26.5	57.3	25.2	54.0	23.3	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.3	9.8	8.6	-10.7	32.7	3.7	1.4	-13.4	33.8	-15.6	34.6	-17.7	35.4	-20.4	36.4
			-11.0	23.4	3.7	1.0	-13.7	24.1	-15.8	24.7	-17.9	25.3	-20.6	26.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	2.6	3.1	5.4	9.0	12.4	15.2	17.6	17.3	14.3	10.5	6.4	3.4
	DBStd	6.39	4.17	4.01	3.38	3.68	3.55	3.10	3.13	2.92	2.72	3.36	3.35	4.25
	HDD10.0	1004	230	194	145	62	15	1	0	0	1	34	116	207
	HDD18.3	3187	488	427	400	279	187	105	54	56	126	243	359	464
	CDD10.0	929	1	1	3	33	89	155	234	227	129	50	6	1
	CDD18.3	71	0	0	0	0	3	9	30	24	4	0	0	0
	CDH23.3	636	0	0	0	8	38	102	264	193	31	1	0	0
	CDH26.7	173	0	0	0	1	4	26	80	59	4	0	0	0
Wind	WSAvg	4.1	4.9	4.7	4.7	4.1	3.9	3.7	3.5	3.3	3.5	4.0	4.1	4.7
Precipitation	PrecAvg	791	65	48	54	48	57	70	78	69	75	69	78	79
	PrecMax	1091	146	132	135	102	144	144	160	158	220	181	129	168
	PrecMin	533	4	4	7	4	16	24	15	6	15	14	20	17
	PrecStd	129	33	28	28	26	29	31	36	36	40	33	27	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	13.1	17.4	24.3	27.0	29.8	32.3	31.3	26.9	21.4	15.5	12.9
		MCWB	10.7	9.4	11.6	15.3	18.1	20.8	20.9	20.7	19.5	16.9	13.5	11.7
	2%	DB	10.5	10.9	14.3	20.8	24.0	26.3	28.9	28.1	23.4	18.4	13.8	11.6
		MCWB	9.4	9.1	10.4	13.3	16.1	19.1	20.4	20.1	18.2	15.6	12.7	10.5
	5%	DB	9.2	9.6	12.1	18.0	21.7	23.5	26.2	25.1	20.9	16.6	12.5	10.3
		MCWB	8.2	8.2	9.3	12.5	15.6	18.0	19.6	19.4	16.9	14.5	11.2	9.4
	10%	DB	8.1	8.5	10.6	15.5	19.2	21.1	23.8	22.8	19.0	15.3	11.1	8.9
		MCWB	7.1	7.3	8.5	11.5	14.5	16.6	18.6	18.1	15.7	13.7	10.1	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	10.6	12.6	16.1	19.7	21.8	22.8	22.5	20.2	17.4	14.1	12.0
		MCDB	11.8	12.0	15.5	22.1	25.1	27.6	28.8	28.2	25.4	20.3	14.7	12.8
	2%	WB	9.6	9.3	11.0	14.3	17.6	19.9	21.3	20.7	18.6	15.9	12.8	10.7
		MCDB	10.3	10.4	13.2	19.1	22.2	24.7	26.7	25.7	22.5	17.9	13.8	11.4
	5%	WB	8.3	8.3	9.8	13.1	16.1	18.5	20.2	19.6	17.4	14.8	11.4	9.4
		MCDB	9.1	9.5	11.7	17.1	20.1	22.4	25.3	24.2	20.3	16.3	12.3	10.3
	10%	WB	7.2	7.4	8.6	11.8	14.7	17.2	19.1	18.6	16.3	13.9	10.1	8.1
		MCDB	8.0	8.4	10.3	15.1	18.6	20.5	23.2	22.2	18.4	15.1	11.0	8.9
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.5	7.2	9.3	9.5	9.4	9.4	9.5	8.6	7.1	5.2	4.6
		MCDBR	4.7	6.3	10.2	13.4	13.3	13.3	13.4	13.0	11.1	8.1	5.6	4.6
		MCWBR	4.4	5.0	6.9	7.5	7.2	7.2	6.3	6.3	6.4	5.6	4.8	4.3
	5% WB	MCDBR	4.6	5.7	9.1	11.9	11.8	11.9	11.9	11.5	10.1	7.7	5.4	4.7
		MCWBR	4.4	4.8	6.5	7.2	7.0	7.0	6.3	6.1	6.1	5.7	4.9	4.6
Clear-Sky Solar Irradiance	taub	0.327	0.346	0.375	0.406	0.405	0.403	0.415	0.408	0.388	0.361	0.340	0.320	
	taud	2.320	2.331	2.268	2.232	2.272	2.295	2.275	2.310	2.360	2.400	2.349	2.310	
	Ebn at Noon	639	749	801	823	844	849	830	813	783	724	617	571	
	Edh at Noon	61	81	105	124	125	124	124	114	96	75	59	52	
All-Sky Solar Radiation	RadAvg	0.61	1.27	2.46	4.07	5.03	5.30	5.04	4.20	3.00	1.70	0.74	0.47	
	RadStd	0.06	0.20	0.28	0.41	0.43	0.39	0.60	0.34	0.31	0.17	0.07	0.05	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page